

DEVICE INVENTORY (15 devices)

Core Routers (3)

Device Model Role Key IPs

Router0 ISR4331 WAN Hub Gi0/0/0: 200.1.2.1
Gi0/0/1: 200.2.3.2

Router6 2911 Site A Gateway Gi0/0: 200.1.2.2
Gi0/1: 192.168.1.1
Gi0/2: 192.168.2.1

Router7 2911 Site B Gateway Gi0/0: 200.2.3.1
Gi0/1: 202.20.20.1
Gi0/2: 201.10.10.1

Switches (4 L2 Switches)

Device Model Ports Role

Switch1 2960-24TT 24 FE + 2 GE Site A VLAN 10 (192.168.1.0/24)

Switch2 2960-24TT 24 FE + 2 GE Site A VLAN 20 (192.168.2.0/24)

Switch3 2960-24TT 24 FE + 2 GE Site B VLAN 30 (202.20.20.0/24)

Switch4 2960-24TT 24 FE + 2 GE Site B VLAN 40 (201.10.10.0/24)

Servers (4)

Device Model IP Subnet DHCP Pool

Server1 Server-PT 192.168.1.3 192.168.1.0/24 serverPool (192.168.1.0/24)

Server2 Server-PT 192.168.2.3 192.168.2.0/24 serverPool (192.168.2.0/24)

Server3 Server-PT 202.20.20.2 202.20.20.0/24 serverPool (202.20.20.0/24)

Server4 Server-PT 201.10.10.2 201.10.10.0/24 serverPool (201.10.10.0/24)

PCs (4)

Device IP Subnet Gateway

PC3 192.168.1.2 192.168.1.0/24 192.168.1.1

PC4 192.168.1.4 192.168.1.0/24 192.168.1.1

PC5 192.168.2.4 192.168.2.0/24 192.168.2.1

PC6 192.168.2.2 192.168.2.0/24 192.168.2.1

DETAILED CONFIGURATION ANALYSIS

ROUTER0 (ISR4331) — WAN Hub

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Internet [FW1: Outside DMZ] [FW2: DMZ Inside] Internal

Interfaces:

Gi0/0/0: 200.1.2.1/24 connects to Router6 (Site A)

Gi0/0/1: 200.2.3.2/24 connects to Router7 (Site B)

Gi0/0/2: Shutdown/unused

Routing (RIP):

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DENY: RFC1918 spoofing (10.0.x.x, 172.16.x.x, 192.168.x.x)

PERMIT: TCP/80 (HTTP)

PERMIT: TCP/443 (HTTPS)

PERMIT: Established TCP connections

DENY: Everything else (implicit)

Propagates WAN subnets to adjacent routers

ROUTER6 (2911) — Site A Gateway

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PERMIT: VLAN 20 (10.0.20.x) Database (10.0.30.x) on port 3306 (MySQL)

PERMIT: VLAN 50 (10.0.50.x) Database (10.0.30.x) on port 3306

DENY: Everything else

Interfaces:

Gi0/0: 200.1.2.2/24 WAN to Router0 (access-list 10 out)

Gi0/1: 192.168.1.1/24 VLAN 10 (PC3, PC4, Server1)

Gi0/2: 192.168.2.1/24 VLAN 20 (PC5, PC6, Server2)

Routing (RIP):

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Core-SW1: standby 1 ip 10.0.10.254, priority 110, preempt

Core-SW2: standby 1 ip 10.0.10.254, priority 100, preempt

ACL 10 (Outbound Filter on Gi0/0):

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ACC-SW1: FE0/3 WEB-BLADE-1 (VLAN 20)

FE0/4 WEB-BLADE-2 (VLAN 20)

ACC-SW2: FE0/3 WEB-BLADE-3 (VLAN 20)

FE0/4 DB-BLADE-1 (VLAN 30)

ACC-SW3: FE0/3 DB-BLADE-2 (VLAN 30)

FE0/4 VIRT-HOST-1 (VLAN 40)

ACC-SW4: FE0/3 VIRT-HOST-2 (VLAN 40)

FE0/4 DNS-DHCP-NTP (VLAN 40)

ACC-SW5: FE0/3 BACKUP-1 (VLAN 70)

FE0/4 BACKUP-2 (VLAN 70)

ACC-SW6: FE0/3 LOAD-BALANCER (VLAN 20)

FE0/4 SOC-SERVER (VLAN 50)

FE0/5 SIEM-SERVER (VLAN 60)

FE0/6 IDS-SENSOR (VLAN 99)

FE0/7 IPS-SENSOR (VLAN 99)

Only allows VLAN 20 (192.168.2.0/24) to exit to WAN

Blocks VLAN 10 (192.168.1.0/24) from leaving the site

ROUTER7 (2911) — Site B Gateway

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Process 1, Area 0 (backbone only)

MD5 Authentication: BANKOSPF2026

Router IDs:

- CORE-R1: 1.1.1.3
- CORE-R2: 2.2.2.3

- CORE-SW1: 3.3.3.1
- CORE-SW2: 3.3.3.2
- AGG-SW1: 5.5.5.1
- AGG-SW2: 5.5.5.2

Network Statement:

- All routers/L3 switches: network 10.0.0.0 0.0.255.255 area 0
- This advertises all 10.0.x.x subnets

Passive Interfaces:

- VLAN SVIs on all switches (don't send hellos to end devices)

Interfaces:

Gi0/0: 200.2.3.1/24 WAN to Router0

Gi0/1: 202.20.20.1/24 VLAN 30 (Server3)

Gi0/2: 201.10.10.1/24 VLAN 40 (Server4)

Routing (RIP):

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AS 65001 (Your Bank)

Neighbors:

- ISP-1 (203.0.113.1, AS 100)
- ISP-2 (203.0.113.5, AS 200)

Router IDs: Same as OSPF (1.1.1.3, 2.2.2.3)

Redistribution:

- Redistribute OSPF routes (so ISPs know how to reach 10.0.x.x)
- Redistribute connected networks (203.0.113.0/29)

Result:

- Your bank is reachable from internet via two ISP links
- Automatic failover if one ISP link fails
- Load balancing across both ISPs

SWITCHES (Switch1, Switch2, Switch3, Switch4)

All four switches use identical baseline configuration:

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Layered Defense (Defense in Depth):

Firewall Layer 1 (FW1):

Denies RFC1918 spoofing

Allows only HTTP/443

Stateful inspection

Firewall Layer 2 (FW2):

Restricts database access

Only VLAN 20/50 30:3306

Blocks lateral movement

VLAN Isolation:

VLAN 20 (web) ≠ VLAN 30 (database)

VLAN 40 (internal) ≠ VLAN 50 (security)

Security by separation

Port Security:

Unused ports shutdown (VLAN 99)

IDS/IPS on separate sensors

SOC monitoring 24/7

Key observations:

Spanning Tree enabled (redundancy-safe)

No VLAN trunking configured

No inter-switch links

No SVI (Switched Virtual Interface) configuration

All ports in default state (effectively access ports)

DATA FLOW & CONNECTIVITY MATRIX

Intra-Site Connectivity

From To Path Status

PC3 (192.168.1.2) Server1 (192.168.1.3) Switch1 L2 only Works (same subnet)

PC5 (192.168.2.2) Server2 (192.168.2.3) Switch2 L2 only Works (same subnet)

PC3 PC5 Router6 (Gi0/1 Gi0/2) Works (RIP routes)

Server3 (202.20.20.2) Server4 (201.10.10.2) Router7 (Gi0/1 Gi0/2) Works (RIP routes)

Inter-Site Connectivity

From To Path Status Notes

PC3 Site A Server3 Site B Router6 Router0 Router7 Switch3 Works RIP advertises routes

Server1 Site A Server4 Site B Router6 Router0 Router7 Switch4 Filtered ACL 10 blocks 192.168.1.0 outbound

PC5 Site A Server4 Site B Router6 Router0 Router7 Switch4 Works ACL 10 permits 192.168.2.0

Server3 Site B Server1 Site A Router7 Router0 Router6 Switch1 Works No return ACL on Router7

Critical Traffic Filtering Asymmetry

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1. Internet packet arrives at FW1
 - Source: 8.8.8.8:1000, Dest: 203.0.113.150:443
2. FW1 ACL checks:
 - RFC1918? No
 - Port 443? Yes
 - PERMIT forward
3. NAT Translation (CORE-R1 or R2):
 - 203.0.113.150 10.0.20.100 (Load-Balancer)
 - Forward to load-balancer
4. VLAN 20 routing:
 - Packet: Dest 10.0.20.100
 - Router: Check routing table
 - Next hop: 10.0.20.1 (CORE-SW1 or SW2 via HSRP)
5. CORE-SW1/2 routing:
 - Destination on same VLAN
 - Forward to access switch (ACC-SW6)
6. Access switch forwarding:
 - Port assignment: Load-Balancer on FE0/3
 - Frame arrives at Load-Balancer
7. Load-Balancer processes:
 - Distributes to WEB-BLADE-1, 2, or 3
 - Web server responds

8. Return path (automatic):

- Reverse routing through same path
- NAT translates back: 10.0.20.100 → 203.0.113.150
- FW1 allows established connection
- Response reaches internet customer

Total latency: 5-20ms depending on device load

This creates one-way communication from Site B to Site A VLAN 10, but not the reverse.

CONFIGURATION SUMMARY TABLE

Aspect	Router0	Router6	Router7
Type	ISR4331	2911	2911
WAN IPs	200.1.2.1	200.2.3.2	200.1.2.2 200.2.3.1
LAN IPs	None	192.168.1.1	192.168.2.1 202.20.20.1
			201.10.10.1
RIP Networks	200.1.2.0	200.2.3.0	192.168.1.0
			192.168.2.0
	200.1.2.0	200.2.3.0	
			201.10.10.0
			202.20.20.0
ACLs	None	ACL 10 (outbound)	None
Spanning Tree	N/A	N/A	N/A
VLAN Config	N/A	L3 routing	L3 routing

DESIGN STRENGTHS & WEAKNESSES

Strengths

RIP dynamic routing — automatically advertises all networks

Simple linear WAN topology — easy to understand and debug

DHCP servers pre-configured — each site can auto-assign IPs

Access list filtering — introduces basic security concept

Weaknesses

RIP is slow — old protocol, high convergence time (30s)

Single WAN path — no redundancy (Router0 is a single point of failure)

Asymmetric ACL filtering — confusing one-way blocks (violates expect-reply principle)

No redundancy at Site A — if Router6 fails, entire site is down

Switches lack SVIs — no Layer 3 at distribution layer

No VLAN trunking — each VLAN requires separate physical switch